

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Indicate which Mathematical Theorem *best fits* the statement.

Fermat's Theorem
Intermediate Value Theorem
Extreme Value Theorem
Mean Value Theorem

- There was a maximum temperature in the last 24 hours.

The Extreme Value Theorem. The temperature outside would be continuous and therefore on the interval from 12am to 12am (the next day) there must exist an absolute maximum temperature.

- A ball tossed in the air will be at its highest point when it stops moving for an instant.

Fermat's Theorem. If the position of the ball is given by a function $s(t)$ then its velocity is given by $s'(t)$. Note that $s'(t)$ exists for all values of t since the velocity of the ball is always defined.

Fermat's theorem then tells us that when $s(t)$ has a local maximum at it must be the case that $s'(t) = 0$.

- A fish that jumped above the surface of the water will make a splash before it is again underwater.

Intermediate Value Theorem. The fish's path through the area will be continuous and since the fish starts beneath the water and can be found outside of the water it must hit the water at some point in between.

- If you travel 90 miles in 2 hours, at some time you were traveling 45 miles per hour.

Mean Value Theorem. A function of the position of the vehicle during these two hours can be drawn and since this function must be both continuous and differentiable the MVT applies and

guarantees there is a value of c within these two hours such that $f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{90 - 0}{2} = 45$

2. Recall that Fermat's Theorem states

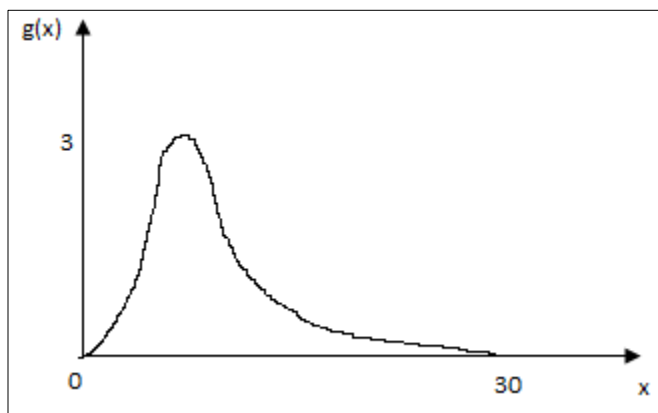
"If f has a local minimum or maximum value at c , and if $f'(c)$ exists, then $f'(c) = 0$."

TRUE OR FALSE: It is also true that this statement can be reverse. That is to say

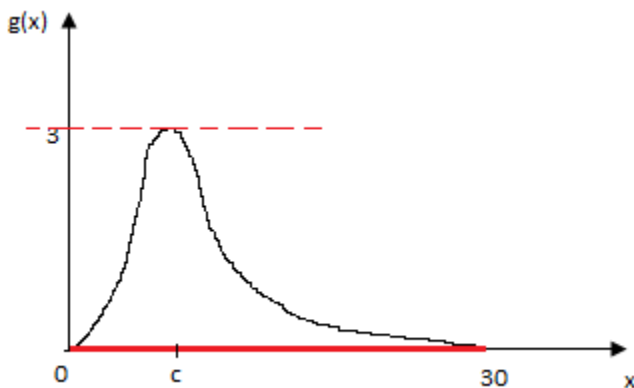
"If $f'(c) = 0$, then f has a local minimum or maximum value at c ."

This is FALSE. We can find several examples where $f'(c) = 0$ but there is neither a local minimum or local maximum at c . For example, consider $f(x) = x^3$ and $c = 0$.

3. A person begins cycling from home at noon. She begins cycling quickly and travels 3 miles north. Then she makes a U-turn without stopping, and then cycles home slowly. She finishes cycling after 30 minutes. Let $g(x)$ be a function that represents the total displacement of the biker from her home versus time.
- a) Construct and label a possible graph of $g(x)$.



- b) Construct a line segment on the graph from part (a) from the starting point to the ending point. On the graph, label the point c guaranteed by the appropriate theorem.



Both Rolle's Theorem and the MVT apply here. Both of these theorem guarantee that there exists a value c in $(0,30)$ such that $g'(c) = 0$

- c) Refer to the cycling context of the problem to describe what the point c means in the problem.

This point is the exact point when the biker turned around. At this moment in time the velocity of the biker was exactly 0.

4. Let $g(x) = x^2 - 4x + 3$.

a) If $a = 0$ and $b = 2$, what value c satisfies the conclusion of the MVT?

We calculate that $g'(x) = 2x - 4$. We see that $g(x)$ is continuous on $[0, 2]$ and differentiable on $(0, 2)$ so the MVT guarantees that there exists a value c in $(0, 2)$ where

$$g'(c) = \frac{g(2) - g(0)}{2 - 0} \rightarrow 2c - 4 = \frac{(-1) - (3)}{2} \rightarrow 2c - 4 = -2 \rightarrow c = 1$$

b) If $a = 0$ and $b = 3$, what value c satisfies the conclusion of the MVT?

We see that $g(x)$ is continuous on $[0, 3]$ and differentiable on $(0, 3)$ so the MVT guarantees that there exists a value c in $(0, 3)$ where

$$g'(c) = \frac{g(3) - g(0)}{3 - 0} \rightarrow 2c - 4 = \frac{(0) - (3)}{3} \rightarrow 2c - 4 = -1 \rightarrow c = \frac{3}{2}$$

c) If $a = 1$ and $b = 3$, what value c satisfies the conclusion of the MVT?

We see that $g(x)$ is continuous on $[1, 3]$ and differentiable on $(1, 3)$ so the MVT guarantees that there

exists a value c in $(1, 3)$ where $g'(c) = \frac{g(3) - g(1)}{3 - 1} \rightarrow 2c - 4 = \frac{(0) - (0)}{2} \rightarrow 2c - 4 = 0 \rightarrow c = 2$

d) In general, what value c do you anticipate satisfies the conclusion of the MVT for any a and b with this polynomial?

I might guess that point directly between the points a and b would satisfy the conclusion of the MVT.

This would be the point $c = \frac{a+b}{2}$.

e) Suppose that $p(x) = Ax^2 + Bx + C$. Show that the number c that satisfies the conclusion of the MVT for any polynomial $p(x)$ is given by $c = \frac{a+b}{2}$

We calculate that $p'(x) = 2Ax + B$. We see that $p(x)$ is continuous on $[a, b]$ and differentiable on (a, b) so the MVT guarantees that there exists a value c in (a, b) where

$$\begin{aligned} p'(c) &= \frac{p(b) - p(a)}{b - a} \rightarrow 2Ac + B = \frac{(Ab^2 + Bb + C) - (Aa^2 + Ba + C)}{b - a} \\ &\rightarrow 2Ac + B = \frac{A(b^2 - a^2) + B(b - a)}{b - a} \rightarrow 2Ac + B = \frac{(b - a)[A(b + a) + B]}{b - a} \\ &\rightarrow 2Ac + B = A(b + a) + B \rightarrow 2c = b + a \\ &\rightarrow c = \frac{b + a}{2} \end{aligned}$$

5. Show that the equation $6x - \cos(x) = 0$ has exactly one real solution.

Show there is a real root

Let $f(x) = 6x - \cos(x)$. We know that f is a continuous because it is a polynomial minus a continuous trig. function. We can see that:

- $f(0) = -1 < 0$
- $f(\pi) = 6\pi + 1 > 0$

Therefore by the IVT there exists a number c in the interval $(0, \pi)$ such that $f(c) = 0$.

Therefore we have shown that $f(x)$ has at least one real root.

Show there cannot be two real roots

Assume that there are two real roots, say $f(c_1) = f(c_2) = 0$ (so we are calling c_1 & c_2 roots).

Since f is continuous everywhere, we know that f is continuous on the interval $[c_1, c_2]$, regardless of what the values of c_1 & c_2 are. It is also differentiable on the interval (c_1, c_2) . Therefore, we have satisfied the conditions for Rolle's Theorem so we can then say, there is a number d in (c_1, c_2) such that $f'(d) = 0$.

We note that: $f'(x) = 6 + \sin(x)$.

However, when we look to find when $f'(x) = 0$ we can see that this is not possible, since $\sin(x) \neq -6$. This is a Contradiction!

Therefore, our initial assumption must be false. There cannot be two real roots.

So, we have shown that the function $f(x) = 6x - \cos(x)$ has exactly one real root.

This implies that the equation $6x - \cos(x) = 0$ has exactly one real solution. ■

6. Suppose f is an odd function and is differentiable everywhere. Prove that for every positive number b , there exists a number c in $(-b, b)$ such that $f'(c) = \frac{f(b)}{b}$.

Since f is differentiable everywhere it is differentiable on $(-b, b)$ and it is also continuous on $[-b, b]$.

Therefore, by the MVT, there exists a value c in $(-b, b)$ such that $f'(c) = \frac{f(b) - f(-b)}{b - (-b)}$

If $f(x)$ is odd, then for all values of x , $f(-x) = -f(x)$. This then implies that

$$f'(c) = \frac{f(b) - f(-b)}{b - (-b)} = \frac{f(b) + f(b)}{b + b} = \frac{2f(b)}{2b} = \frac{f(b)}{b}.$$

7. Suppose that f and g are continuous on $[a, b]$ and differentiable on (a, b) . Suppose also that $f(a) = g(a)$ and $f'(x) < g'(x)$ for $a < x < b$. Prove that $f(b) < g(b)$.

Consider the function $h(x) = f(x) - g(x)$. Note that since f and g are continuous on $[a, b]$ and differentiable on (a, b) , so is h .

Therefore, by the MVT, there is a value c in (a, b) such that $h'(c) = \frac{h(b) - h(a)}{b - a}$.

We then see that if $h'(c) = \frac{h(b) - h(a)}{b - a}$ then

$$f'(c) - g'(c) = \frac{[f(b) - g(b)] - [f(a) - g(a)]}{b - a} = \frac{f(b) - g(b)}{b - a}$$

Since it is given that $f'(x) < g'(x)$ then it must be the case that $f'(c) - g'(c) < 0$. However, this then

implies that $\frac{f(b) - g(b)}{b - a} < 0 \rightarrow f(b) - g(b) < 0 \rightarrow f(b) < g(b)$.